

Schizophrenia

Myths and facts?

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1. Psychiatric disorders are **not true medical illnesses** like heart disease and diabetes.

❖ **Fact:** Brain disorders, like heart disease and diabetes

2. People with a severe mental illness, such as schizophrenia, are usually **dangerous and violent**.

3. Mental illness is the result of bad parenting

❖ **Fact:** Mental illness has a physical cause (**changes in brain chemistry**) and is not the result of bad parenting.

Introduction

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- Schizophrenia is a **chronic heterogeneous syndrome** of disorganized & bizarre(unusual) thoughts, delusions, hallucinations, inappropriate affect, cognitive deficits & impaired psychosocial functioning

Epidemiology

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- It is the **4th leading cause of disability** among adults
- It **affects 1 % of the world's population**
- Schizophrenia **equally affect all race, genders & culture.**
- Approximately **10% of people with schizophrenia die by suicide.**
- Symptoms usually **first present in late adolescence or early adulthood**

Male: b/n 15 to 24 yrs old

Female: b/n 25 to 34 yrs old

Pathophysiology

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- **Multiple etiologies likely exist:**
- *There is strong evidence supporting a genetic basis for the disorder.*
 - ▣ *First-degree relatives carry 10% risk*
 - ▣ *Identical twins have risk of 50%.*

Dopaminergic hypothesis:

📖 *Psychosis may result from **hyper-** or **hypoactivity** of dopaminergic processes in specific brain regions*

- **Positive symptoms: DA receptor hyperactivity** in the mesolimbic DA pathway

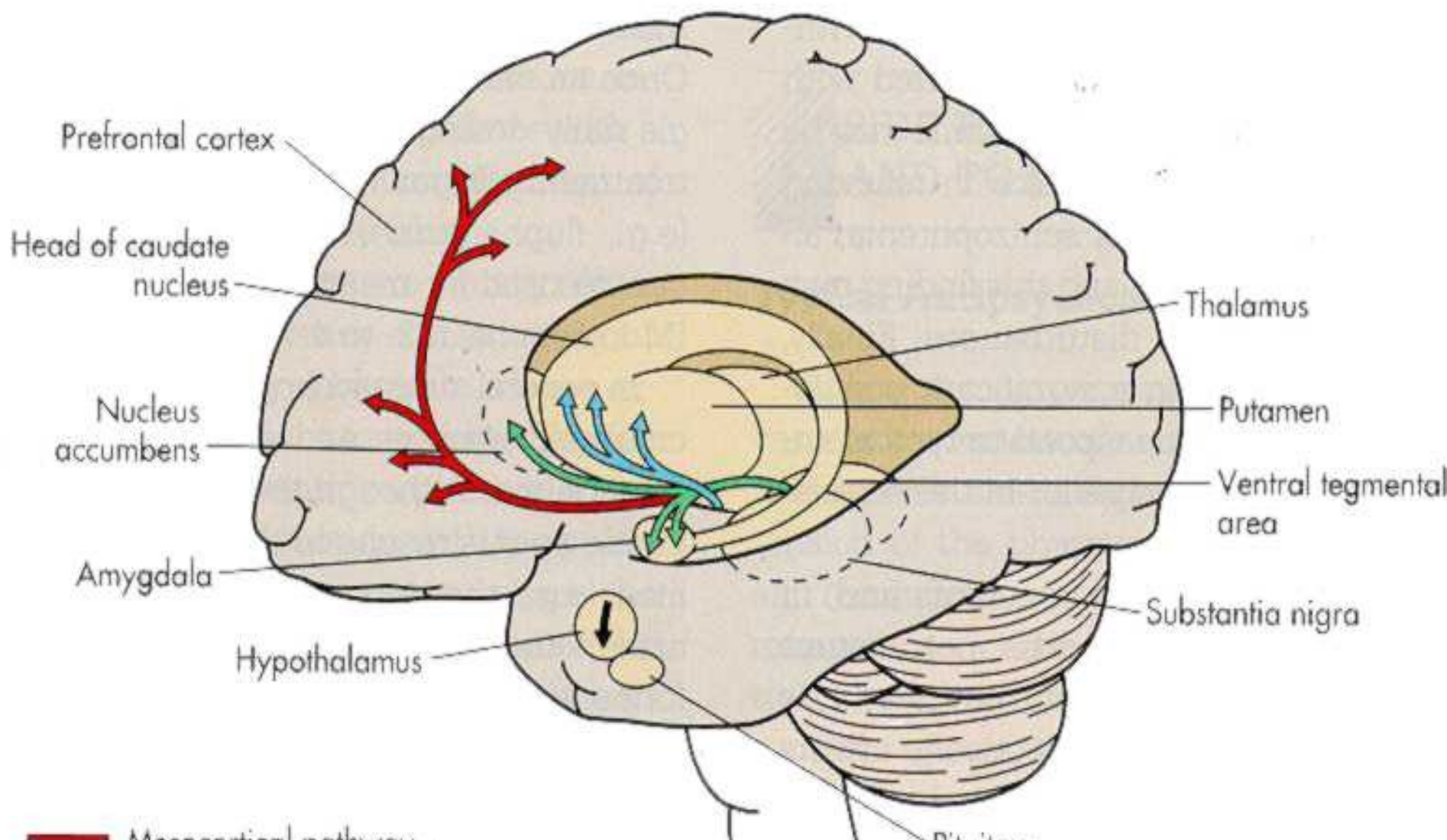
- **Negative and cognitive symptoms:** DA receptor **hypofunction** in the prefrontal cortex (mesocortical)

Glutamatergic dysfunction:

- ✚ A **deficiency of glutamatergic** activity produces symptoms similar to those of **dopaminergic hyperactivity** and possibly symptoms seen in schizophrenia

Serotonin (5-HT) abnormalities:

- ✚ Schizophrenic pts with abnormal brain scans have **higher** whole blood **5-HT_{2A} conc.**



- Mesocortical pathway
- Mesolimbic pathway
- Nigrostriatal pathway
- Tuberoinfundibular pathway

- Blockage.....leads to desirable
- Leads to EPS
- **induces endocrine disorders**

Clinical Symptoms

I. Positive (in excess) 4 major symptoms.....

□ A. Delusion

- Fixed false belief.....mistaken perceptions or experiences

- Suspiciousness (paranoid).....most common

- e.g. believing that someone is watching them through hidden cameras

- People are out there to harm them

- In severe cases.....FBI plot (of control e.g. believing that an alien force is controlling their actions)

Clinical Symptoms

I. Positive (in excess) 4 major symptoms.....

B. Hallucination: (seeing, hearing, or feeling things that aren't really there)

All 5 senses.....but

- ▣ Hearing voices.....Auditory – **most common** form
- ▣ See things.....
- ▣ smell bad odor.....
- ▣ Voice giving them instruction.....

Clinical Symptoms

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I. Positive (in excess)

C. Disorganized behavior

- ▣ wearing double in hot weather
- ▣ It can include odd, bizarre behaviour such as smiling, laughing, or talking to oneself or being preoccupied/responding to internal stimuli

D. Disorganized speech: loose speech

- ▣ Jump topic to topic
- ▣ Hard to understand
- ▣ No logical order

E. Catatonic behavior

-movement and response (agitation)

2. Negative symptoms:

- Qualities 'taken away' from the personality
 - Impoverished(make poor) speech (**alogia**) and thinking,
 - Lack of social drive (**asociality**)
 - **Flatness** of emotional expression, and
 - Apathy(lack of interest): **Anhedonia**

3.Cognitive symptoms:

- Impaired attention, working memory, and executive function

Features of Schizophrenia

Positive Symptoms

Delusions
Hallucinations
Disorganized speech
Catatonia

Negative Symptoms

Affective flattening
Alogia
Avolition
Anhedonia
Social withdrawal

Social/occupational Dysfunction

Work
Interpersonal relationships
Self-care

Cognitive Deficits

Attention
Memory
Executive functions
(e.g., abstraction)

Mood Symptoms

Depression
Anxiety
Hopelessness
Demoralization
Stigmatization
Suicidality

Comorbid Substance Abuse

□ **DSM-5**, specifies the following criteria for diagnosis of schizophrenia:

- ✓ Two or more symptoms **(at least 1 month)**, including hallucinations, delusions, disorganized speech, grossly disorganized or catatonic behavior & negative symptoms
- ✓ Significantly impaired functioning (work, interpersonal, or self-care)

- *There is **no objective** diagnostic or laboratory measure or brain imaging technique to diagnose schizophrenia.*
- **A thorough mental status examination:**
 - Physical and neurologic examination
 - Complete family and social history
 - Vital signs and laboratory workup
 - CBC, hepatic function, renal function, ECG, fasting serum glucose, serum lipids, thyroid function, and urine drug screen

Desired Outcome:

- **The goals of treatment include the following:**
 - ✓ Alleviation of target symptoms
 - ✓ Improvement in psychosocial functioning and productivity
 - ✓ Avoidance of side effects
 - ✓ Compliance with the prescribed regimen

□ **Typical antipsychotics (FGA)**

- Bind to **D2** (in Mesolimbic structure)
- **Low potency:** Chlopromazine, thioridazine
- **High potency:** Haloperidol, thiothixene, fluphenazine

□ **Atypical antipsychotics (SGA)**

- Clozapine Risperidone
- Olanzapine Quetiapine
- Ziprasidone Sertindole
- Aripiprazole

- All current SGAs have a greater affinity for 5-HT_{2A} receptors than D₂ receptors
- First choice in treatment of schizophrenia (except **clozapine**)
- Have **superior efficacy** for treatment of **negative, cognitive and mood** symptoms.
- Cause few or no acutely occurring **EPS** & tardive dyskinesia(**TD**).
- Have less effect on serum prolactin.
- Clozapine is used for treatment resistant schizophrenia & suicidal ideation cases

Antipsychotics

	EPS/TD	Prolactin	Efficacy
1st Generation Haloperidol, chlorpromazine	High	Elevation	Positive Symptoms
2nd Generation Risperidone, Ziprasidone	Dose- Dependant	Elevation	Positive and Negative symptoms
3rd Generation Clozapine, Olanzapine, Quetiapine	Low	Sparing	Positive, Negative, Cognition, and Mood Symptoms

Selection of AP should be based on:

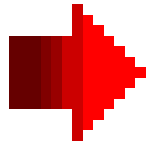
- (1) The need to avoid certain side effects,
- (2) Concurrent medical or psychiatric disorders, and
- (3) Pt or family history of response
- (4) Availability and cost

NB: Negative symptoms are generally **less responsive** to antipsychotic therapy

Initial Therapy

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- The goals during the **first 7 days** are:



Decreased agitation, hostility, anxiety, aggression and normalization of sleep and eating patterns.

- Titrate dose over the first few days to an average effective dose.

- ***If there is no improvement within 3 to 4 wks at therapeutic doses, then an alternative AP should be considered***

- In general, rapid titration of **AP** dose is not recommended

Stabilization Therapy

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- During wks 2 and 3, goals should be **to improve socialization, self-care habits, and mood.**
- Improvement in formal thought disorder
 - ▣ may require an additional 6 to 8 wks.
- Dose titration may continue every 1 to 2 wks as long as the pt has no side effects
- If symptom improvement is not satisfactory after 8 to 12 wks, a different strategy should be tried

Maintenance Therapy

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- Medication should be continued for at ***least 12 mths*** after **remission** of the first psychotic episode.
 - ▣ Continuous treatment is necessary in most pts at the lowest effective dose
- AP (especially ***FGAs & clozapine***) should be **tapered slowly** before d/c to avoid ***rebound cholinergic withdrawal symptoms***
- In general, when switching from one **AP** to another, the first should be tapered and d/c over 1 to 2 wks after the second AP is initiated.

Depot Antipsychotic Medications

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- **Risperidone** : is first SGA as long acting injectable
 - 25-50mg IM every 2 wk
- **Fluphenazine decanoate:**
 - 1.2 -1.6 times oral daily dose (rounding up to nearest **12.5 mg**) **weekly**
- **Haloperidol decanoate:**
 - 10-15 times oral daily dose (rounding up to nearest 50mg) every month

□ **Principle for conversion from oral to depot formulation is**

Stabilize on an oral dosage form of the same agent (or at least a short trial of 3-7 days) to be sure the medication is tolerated adequately.

Augmentation therapy

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- Involves the addition of a non-antipsychotic drug to an AP in a poorly responsive pt.
- **Mood stabilizers** (e.g., **lithium**, **valproic acid**, and **carbamazepine**) used as augmentation agents may **improve labile affect** and **agitated behavior**
- Responders to augmentation therapy usually improve rapidly.
- If there is no improvement, the augmenting agent should be discontinued.

- **SSRI** have been used with FGAs with improvement of **negative symptoms**.
- **Propranolol, pindolol, and nadolol** have been used for **anti-aggressive effects**

Combination

- Combining FGA and SGA or combining d/nt SGAs have been suggested, but no data exist to support or refute these strategies.

Dosage regimen of antipsychotic medications

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Medication	Optimal Dose	Maximum Dose
amisulpride*	537 mg/day	1,200 mg/day
aripiprazole oral	11.5 mg/day	30 mg/day
aripiprazole LAI	463 mg/4 weeks	882 mg/4 weeks
asenapine	15.0 mg/day	20 mg/day
brexpiprazole	3.36 mg/day	4 mg/day
cariprazine	7.6 mg/day	6 mg/day
clozapine	567 mg/day	600 mg/day
haloperidol oral	6.3 mg/day	20 mg/day
iloperidone	20.13 mg/day	24 mg/day
lurasidone	147 mg/day	160 mg/day
olanzapine oral*	15.1 mg/day	20 mg/day
olanzapine LAI	277 mg/2 weeks	300 mg/2 weeks
paliperidone oral	13.4 mg/day	12 mg/day
paliperidone LAI	120 mg/4 weeks	234 mg/4 weeks
quetiapine	482 mg/day	800 mg/day
risperidone oral	6.3 mg/day	16 mg/day
risperidone LAI	36.6 mg/2 weeks	50 mg/2 weeks
sertindole	22.5 mg/day	24 mg/day
ziprasidone	186 mg/day	200 mg/day

Management of Side Effects of AP

Anticholinergic Side Effects

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- Include impaired memory, dry mouth, constipation, tachycardia, blurred vision, inhibition of ejaculation, and urinary retention.
 - ▣ Low potency **FGAs**, **clozapine**, and **olanzapine** are most likely to cause
 - ▣ Elderly pts are especially sensitive
- **Dry mouth**: increased intake of fluids, oral lubricants
- **Constipation**: increases in exercise, fluid, and dietary fiber intake

i. Dystonia

- Dystonias are prolonged **tonic muscle contractions**, with rapid onset (usually within 24 to 96 hrs of dosage initiation or dosage increase); they may be *life threatening*
- Occur primarily with **high-potency FGAs**.
- RX: IM anticholinergic, diazepam
- Oral anticholinergic for prevention

Dystonia



ii. Pseudoparkinsonism

☐ Symptoms may include:

- (1) ***Akinesia***, bradykinesia, or decreased motor activity
- (2) **tremor** (at rest and decreasing with movement)
- (3) **rigidity**, which may present as stiffness
- (4) Postural abnormalities

☐ ***Risk factors are FGAs (high dose), increasing age, and possibly female gender.***

- The onset of symptoms is usually 1 to 2 wks after initiation or dose increase

◆ Management:

- Benztropine, **Trihexyphenidyl**, diphenhydramine,
- ***Amantadine is as efficacious as AChs and has less effect on memory.***

iii. Akathesia

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- somatic restlessness and inability to stay still or calm
- Management:
 - Reduce the dose of antipsychotic
 - Change to a lower-risk antipsychotic
 - Beta-blockers – propranolol

iv. Tardive Dyskinesia (TD)

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- TD is **irreversible** and is characterized by **abnormal involuntary movements** occurring with chronic antipsychotic therapy.
- Classic presentation is **buccolingual-masticatory movements (BLM)**.
 - ▣ Facial movements include frequent blinking, grimacing, upward deviation of the eyes, and lip smacking.
 - ▶ Dosage reduction or discontinuation
 - ▶ Switching the pt with TD to **clozapine** is a first-line strategy
 - ▶ Valbenazin and deutetrabenazine

TD

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- Administration of most or all of the daily dose at bedtime can decrease daytime sedation
- The **SGAs** as first-line treatment have been shown to **improve cognition** (attention to tasks and improved working memory) over a **9-mth period**.

- The highest risk for antipsychotic-induced seizures is with the use of **chlorpromazine** or **clozapine**.
- **Seizures** are more likely with initiation of treatment and with the use of higher doses and rapid dose increases.
- When an isolated seizure occurs, **a dosage decrease** is recommended
 - ▣ Anticonvulsant use is not recommended

Endocrine System

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- **Elevations in prolactin** levels (hyperprolactinemia) result in gynecomastia, galactorrhea, menstrual irregularities, decreased libido, and sexual dysfunction
 - More common with the **use of FGAs and risperidone**.
 - Possible management strategies for galactorrhea include switching to an **SGA** (e.g., olanzapine, quetiapine, aripiprazole, or ziprasidone).
- **Weight gain** is frequent with AP therapy including SGAs, especially **olanzapine and clozapine**
- **Type 2 diabetes:** Schizophrenics have a higher prevalence
 - ▣ Clozapine and olanzapine may be more likely, and aripiprazole and ziprasidone may be less likely to cause this

- The incidence of **orthostatic hypotension** (> 20 -mm Hg drop in SBP upon standing) is greatest with **low-potency FGAs**, especially with IM or IV administration.
 - ▣ Tolerance to this effect usually occurs within 2 to 3 mths.
 - ▣ Reducing dose or changing to an AP with less α -adrenergic blockade
- **Thioridazine, clozapine & ziprasidone** are more likely to cause **ECG changes**
- In pts older than 50 years, pretreatment ECG and **serum K** and **Mg levels** are recommended.

- Some **SGAs** and ***phenothiazines*** cause elevations in serum triglycerides and cholesterol.
- The risk for this effect may be ***less*** with risperidone, ziprasidone, and aripiprazole

Psychiatric side effects

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- ***Include:*** apathy and withdrawal, depression
- Delirium & psychosis may occur with high doses of FGA and combination of FGA and anticholinergics

□ **Hepatic system**

- LFT abnormalities are common (elevation of aminotransferases)
- **Cholestatic hepatitis** has been reported with **risperidone**

□ **Genitourinary system**

- Urinary retention with low potency FGA
- Urinary incontinence is especially problematic with **clozapine**
- **Sexual dysfunction:** FGA and Risperidone

- **Transient leukopenia** may occur with AP therapy
 - If WBC count is less than 3,000/mm³ or ANC is less than 1,000/mm³, AP should be d/c, and WBC count monitored closely until it returns to normal.
- **Agranulocytosis** reportedly occurs in 0.01% of pts receiving FGAs, and it may occur more frequently with **chlorpromazine and thioridazine**.
- 1% in clozapine, required frequent checkup

Relative Side-Effect Incidence of Commonly Used AP

	Sedation	EPS	Anticholinergic	Orthostasis	Weight Gain	Prolactin
Aripiprazole	+	+	+	+	+	+
Chlorpromazine	++++	+++	+++	++++	++	+++
Clozapine	++++	+	++++	++++	++++	+
Fluphenazine	+	++++	+	+	+	++++
Haloperidol	+	++++	+	+	+	++++
Olanzapine	++	++	++	++	++++	+
Perphenazine	++	++++	++	+	+	++++
Quetiapine	++	+	+	++	++	+
Risperidone	+	++	+	++	++	++++
Thioridazine	++++	+++	++++	++++	+	+++
Thiothixene	+	++++	+	+	+	++++
Ziprasidone	++	++	+	+	+	+

Evaluation of Therapeutic Outcomes

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- Patient-rated **self-assessments** can also be useful
- **Weight** should be monitored monthly for 3 mths, then quarterly.
- **Body mass index**, waist circumference, blood pressure, fasting plasma glucose, and fasting lipid profile should be monitored at the end of 3 mths, then annually.

Case

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CC: “I want to see my lawyer.”

HPI: This is the first admission for AG, a 32-year-old woman who was brought to the state hospital by the police. The patient apparently has been *delusional and believes people sneak* into her room at night when she is asleep and place a thousand worms inside her body. She also believes that she is *being raped* by passing men on the street. She is quite preoccupied about *having massive wealth*. She claims to have bought some gold and left it at the grocery store. She believes that her ideas have been given to a *government* who has had plastic surgery to look like her and is using her ID to take possession of all of her property.

- Apparently, precipitating event causing her hospitalization was that she created a disturbance at a local fast-food restaurant, claiming that she owned it. Because of the disturbance, police were called, and she subsequently was sent here on an order of protective custody. She claims that 6 yrs ago she was raped by a relative and broke her hip in the process. She states that her feet were cut off because she would not do what her impostors wanted her to do, and her feet were subsequently sent back to her from Central America and were reattached.

- Her speech is quite rambling. She speaks of having been part of an experiment in which 38 eggs *were taken from her body*, and children were produced from them and then killed by the government. She claims that she has *worms in her* that are the type that kill dogs and horses and says that they have been put there by the government. She also claims that at one time she had transmitters in her backbone and that it took 3 yrs to have them taken out by the government.

1. Which information (symptoms) indicates the presence or severity of an acute exacerbation of schizophrenia, paranoid type?
 - ▣ Delusion and paranoid type
 - ▣ Grandiose ideation
 - ▣ Disorganized speech

Desired Outcome

2. What are the goals of pharmacotherapy in this case?

- *Achieve and then maintain control of the target symptoms of schizophrenia.*
- *Promote patient adherence to the prescribed regimen.*
- *Minimize side effects from medications.*
- *Improve the **patients** quality of life so that she can function as a contributing member of society.*
- *Provide effective treatment that **will** decrease **long-term** hospitalizations and perhaps **avoid suicide**.*

Therapeutic Alternatives

3.a. What nondrug therapies might be useful for this patient?

- ▣ Medication education classes to promote compliance and understanding about the illness
- ▣ Group therapy for support one symptoms are under control with medication
- ▣ Vocational training to promote independence

3.b. What pharmacotherapeutic options are available for the treatment of this patient?

- ▣ **1st generation (typical AP):**
- ▣ **2nd generation Atypical AP:**
- ▣ **Selection criteria ?????**

- What drug, dosage form, dose and schedule of therapy are best for this patient?
 - ▣ *Atypical AP drugs are optimal.*
 - ▣ *Other atypical AP drugs (Olanzapine)*
- What alternatives would be appropriate if the initial therapy fails or cannot be used?
 - ▣ *The pt could be started on an other atypical AP. E.g. if she received risperidone initially, a trial of aripiperazole, ziprazidone, quetiapine can be started.*
- What are the differences between FGA vs SGA?

THE END!!!